



Jain College of Engineering, Belagavi

Dept. of Electronics and Communication Engineering



Sub Code(BEC586)

**Topic Name: Contactless Laser and Remote Control Switch Using
TSOP Sensor for Physically Abled Persons**

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AIM:

Design & Implement of Contactless Laser and Remote Control Switch Using TSOP Sensor for Physically Abled Persons

Introduction:

- ❑ In today's world, automation and contactless control systems play a crucial role in improving comfort, hygiene, and accessibility.
- ❑ The contactless laser and remote control switch using a TSOP sensor is designed to operate electrical appliances without physical touch.

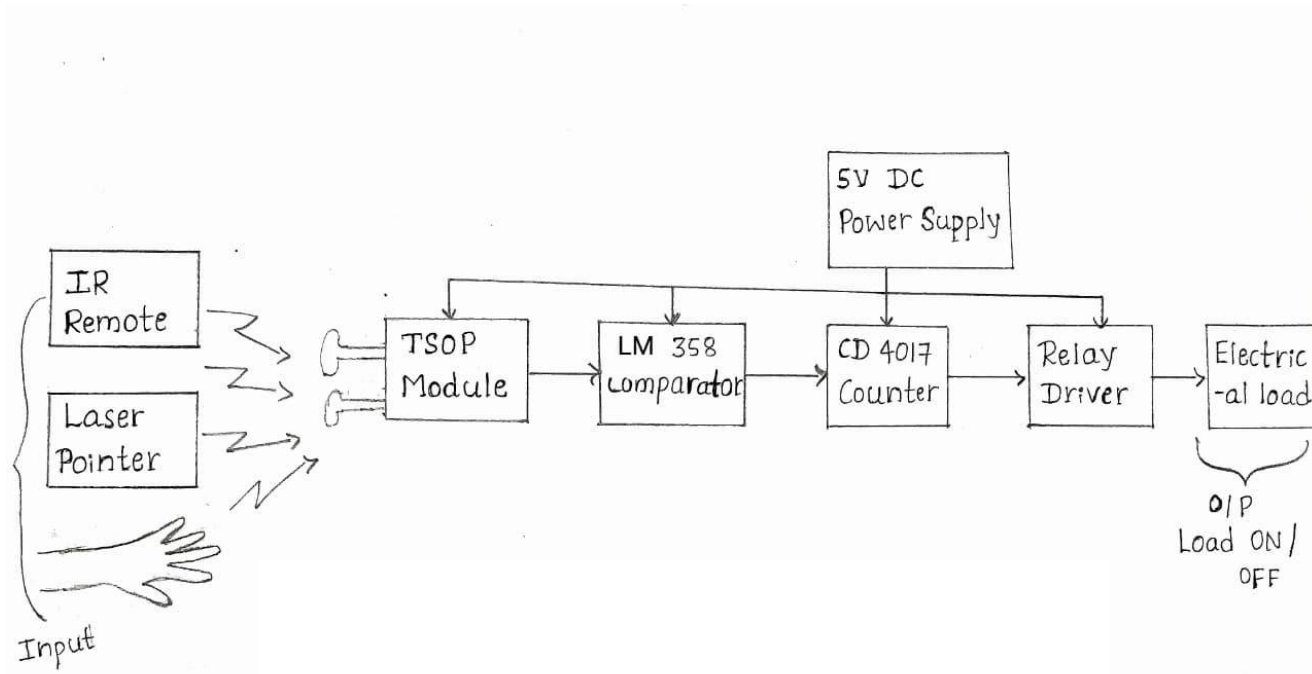
Abstract

- ❑ The **“Contactless Laser and Remote Control Switch Using TSOP Sensor for Physically Abled Persons”** project aims to provide a convenient, hygienic, and efficient method of controlling electrical appliances without physical contact. The system integrates a TSOP (IR receiver) sensor with a laser pointer and a remote control mechanism to detect user commands and trigger the switching of devices. By eliminating the need to touch switches, it reduces the risk of contamination, improves accessibility, and enhances convenience in daily life.
- ❑ This project is particularly beneficial for abled persons who seek smart home automation solutions and a modernized approach to controlling electrical appliances.

Literature Survey

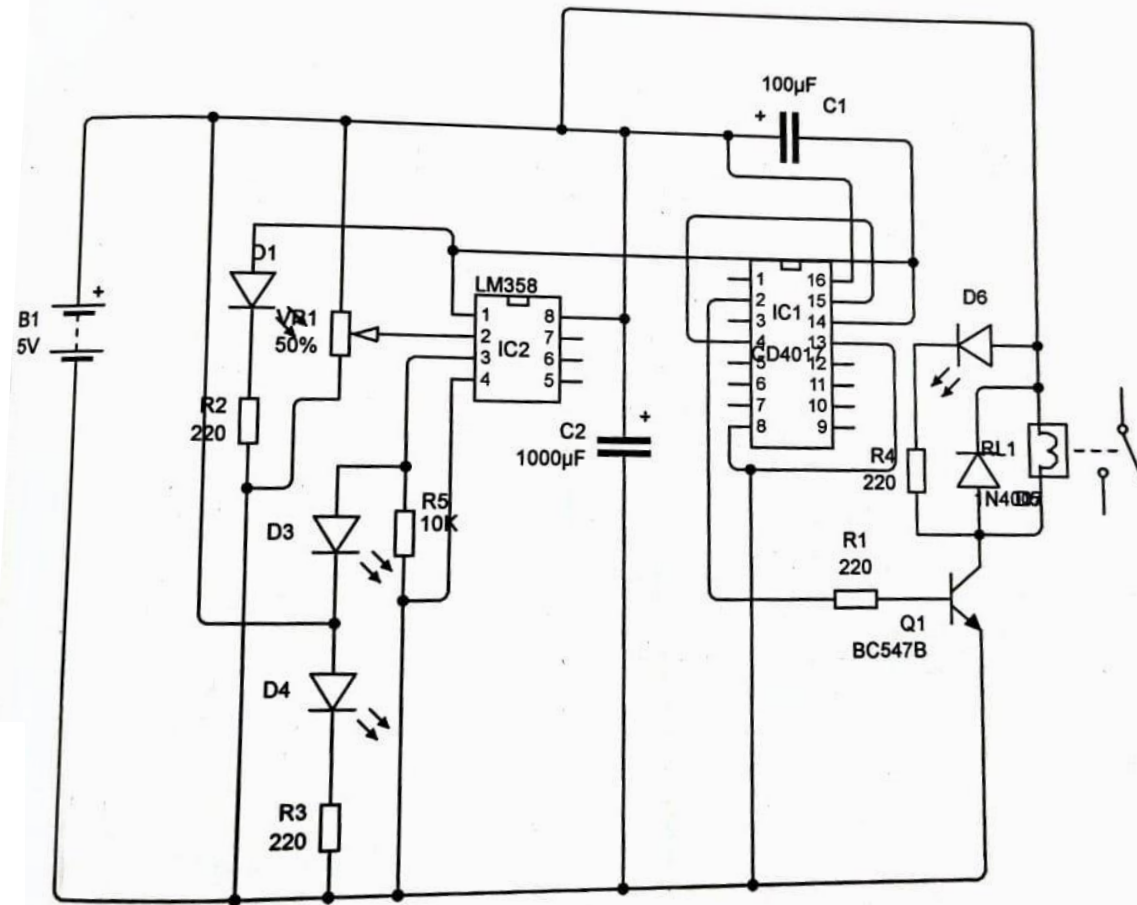
AUTHOR NAME	YEAR	TITLE	RESULT AND CONCLUSION
Sriram Ramachandran, Prashant R Nair, Shriram KVasudevan	2018	Cane Free Obstacle Detection using Sensors and navigational guidance	The project concentrates on obstacle detection for visually impaired people using IR and TSOP
Cengiz, M Azarifar M Arik	2022	A critical review on the junction temperature measurement of light emitting diodes	light emitting diodes (LEDs) have been proven to be the most efficient to conventional light source
Lakshmi Boppana, Vishal Jain, Ravi Kishore	2019	Smart stick for elderly	This walking stick can monitor various health parameters and caretaker of any abnormalities.
Mr Bhushan Nemade	2023	Design and Development of Smart Warehouse Robot Prototype	The use of robotics in warehouse management can also help reduce labor costs and minimize human errors.

BLOCK DIAGRAM



**Contactless Laser and Remote Control Switch Using
TSOP Sensor for Physically Abled Persons**

CIRCUIT DIAGRAM



Specification: Hardware Components

1. IC LM 358

Dual op-amp ,used for analog applications, filters oscillators etc.

2.IC CD 4017

Commonly used in LED chasers, digital counters, timing circuits, and sequential switching

3.Tsop sensor module

Tsop sensor are IR receives modules.

4.Relay(5v\10vAmp)

Electrically operated switch that uses a low power signal to control a separate ,higher power ckt.

5.Trimmer

Trimmer is a small adjustable register used to calibrate.

Advantages

1. Hygienic Operation: Since the switch is contactless, there is no need to physically touch the device, reducing the risk of spreading germs and bacteria. Useful in public areas or for users conscious about cleanliness.

2. Ease of Use: The system can be operated without physically pressing buttons, making it convenient and fast.

Even physically abled users benefit from less effort and more comfort in operation.

3. Increased Device Lifespan: No mechanical contact means fewer wear-and-tear issues on the switch, increasing the longevity of the device.

4. Remote Accessibility: Using TSOP sensors with a remote allows operation from a distance, improving convenience in situations where reaching the switch is cumbersome.

5. Safety: Contactless operation reduces the risk of electrical shocks or short circuits caused by direct handling.

Applications

- 1.Wear and Tear:** Mechanical switches have moving parts that can wear out overtime due to repeated operation, leading to failure.
- 2.Arcing:** When switching high-current loads, mechanical switches can produce electrical arcs, which can degrade contacts and pose safety hazards.
- 3.Limited Lifespan:** They have a finite number of operational cycles before failure, making them less durable compared to electronic alternatives.
- 4.Slow Response:** Mechanical switches operate slower than solid-state or electronic switches, which can be critical in some applications.
- 5.Manual Operation:** Most mechanical switches require manual operation, which may not be convenient for automation or remote control.

Reference

- ❑ Vaibhav, S, et al, “Smart' Cane for the Visually Impaired: Design and Controlled Field Testing of an Affordable Obstacle Detection System", International Conference on Mobility and Transport for Elderly and Disabled Persons. 2010.
- ❑ Gaikwad, A. G., & Waghmare, H. K, “Smart Cane Indicating a Safe free Path to Blind People Using Ultrasonic Sensor”, International Journal on Recent and Innovation Trends in Computing and Communication, Feb. 2016, Volume 4, Issue 2, pp.179-183
- ❑ Sakhardande, I., Pattanayak, P., & Bhowmick, M, “Smart Cane Assisted Mobility for the Visually Impaired”, International Journal of Computer, Electrical, Automation, Control and Information Engineering Vol:6, No:10, 2012, pp. 1262-1265.].Clerk Maxwell, A Treatise on Electricity and Magnetism, 3rd ed., vol. 2. Oxford:Clarendon, 1892, pp.68-73

Conclusion

- ❑ The “**Contactless Laser and Remote Control Switch Using TSOP Sensor**” successfully demonstrates a convenient and hygienic method of controlling electrical appliances without physical touch. The system ensures ease of use, improves accessibility, and offers a modern solution for smart home automation. Its reliable and cost-effective design makes it suitable for practical implementation in homes, offices, and public spaces. Overall, this project highlights how simple technology integration can enhance comfort, safety, and efficiency in daily life.

THANK YOU